

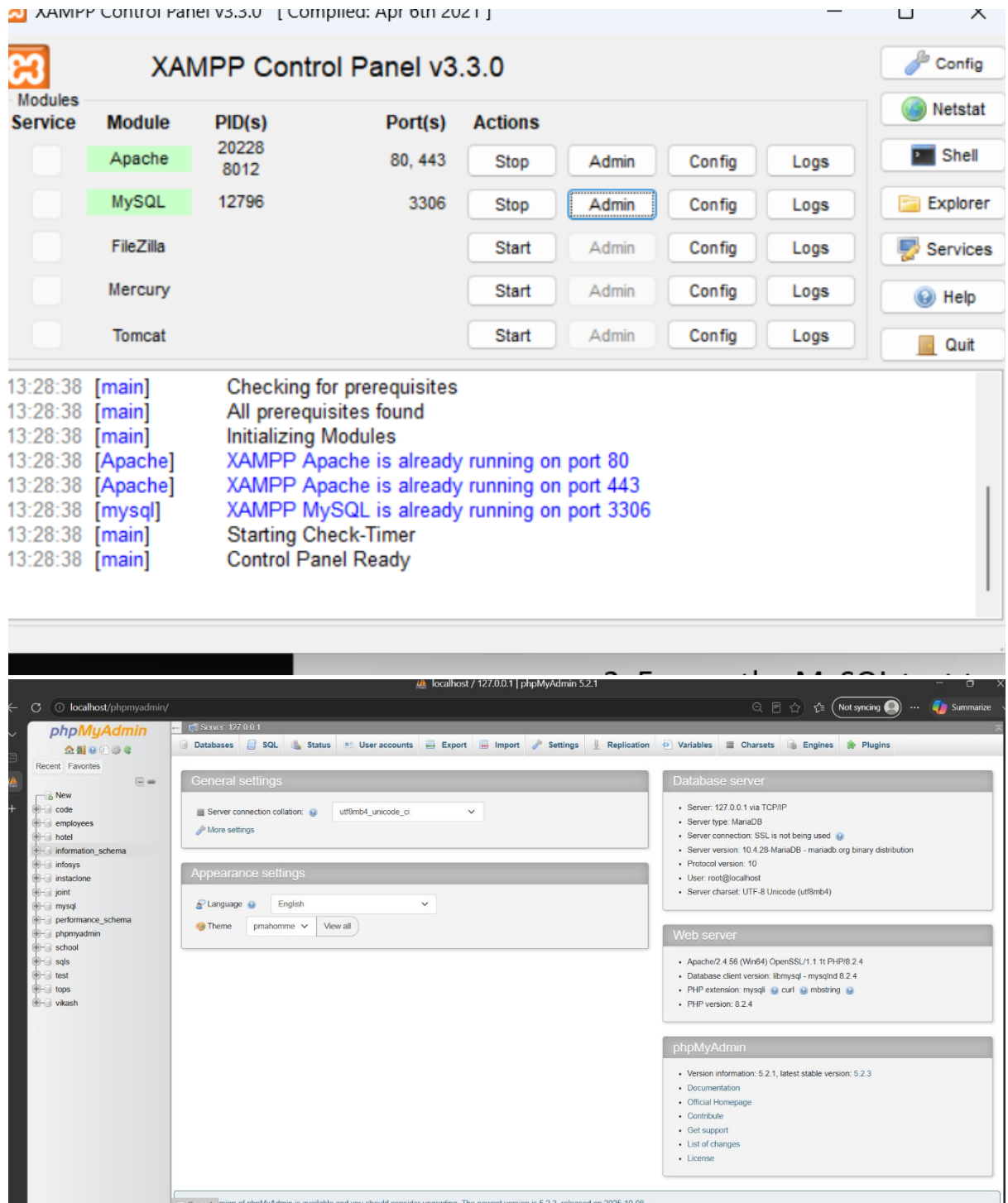
Session-1

1. Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.

Ans.

1. Open the xampp control panel on your computer.
2. Click the start button next to mysql (and apache if needed).
3. Ensure the MySQL text turns green, indicating it is running successfully.

I am using the xampp control panels show the screen shorts



2. Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

Ans.

The screenshot shows the phpMyAdmin interface. The top navigation bar includes 'Databases', 'SQL', 'Status', 'User accounts', and 'Export'. A dropdown menu is open for 'Run SQL query/queries on server "127.0.0.1":', showing the SQL command: `1 CREATE DATABASE InstaClone;`

Below this, the interface shows the 'Database: instaclone' view. The top navigation bar includes 'Structure', 'SQL', 'Search', 'Query', 'Export', 'Import', 'Operations', 'Privileges', 'Routines', 'Events', and 'Triggers'. A dropdown menu is open for 'Run SQL query/queries on database instaclone:', showing the SQL command: `1 CREATE TABLE Users(user_id int PRIMARY KEY AUTO_INCREMENT,username varchar(30),email varchar(100),followers_count int);`

Below this, the interface shows the 'Query results' view. The top navigation bar includes 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', and 'Import'. A green message box states: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)'. Below this, the SQL command `SELECT * FROM `users`` is shown. A checkbox for 'Profiling' is checked, with links for 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. Below this, the table structure is shown with columns: `user_id`, `username`, `email`, and `followers_count`. A dropdown menu is open for 'Query results operations', showing a 'Create view' button.

3. Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

Ans.

View query results

✓ 3 rows inserted.
Inserted row id: 3 (Query took 0.0101 seconds.)

INSERT INTO users(username,email,followers_count)VALUES('pandeyvsp','vp11@gmail.com',1900), ('karan123','karan@gmail.com',2300), ('ni_bapu','nm@gmail.com',23);

[Edit inline](#) [[Edit](#)] [[Create PHP code](#)]

✓ Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

SELECT * FROM `users`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				user_id	username	email	followers_count
☐				1	pandeyvsp	vp11@gmail.com	1900
☐				2	karan123	karan@gmail.com	2300
☐				3	ni_bapu	nm@gmail.com	23

☐ Check all | With selected: Edit Copy Delete Export

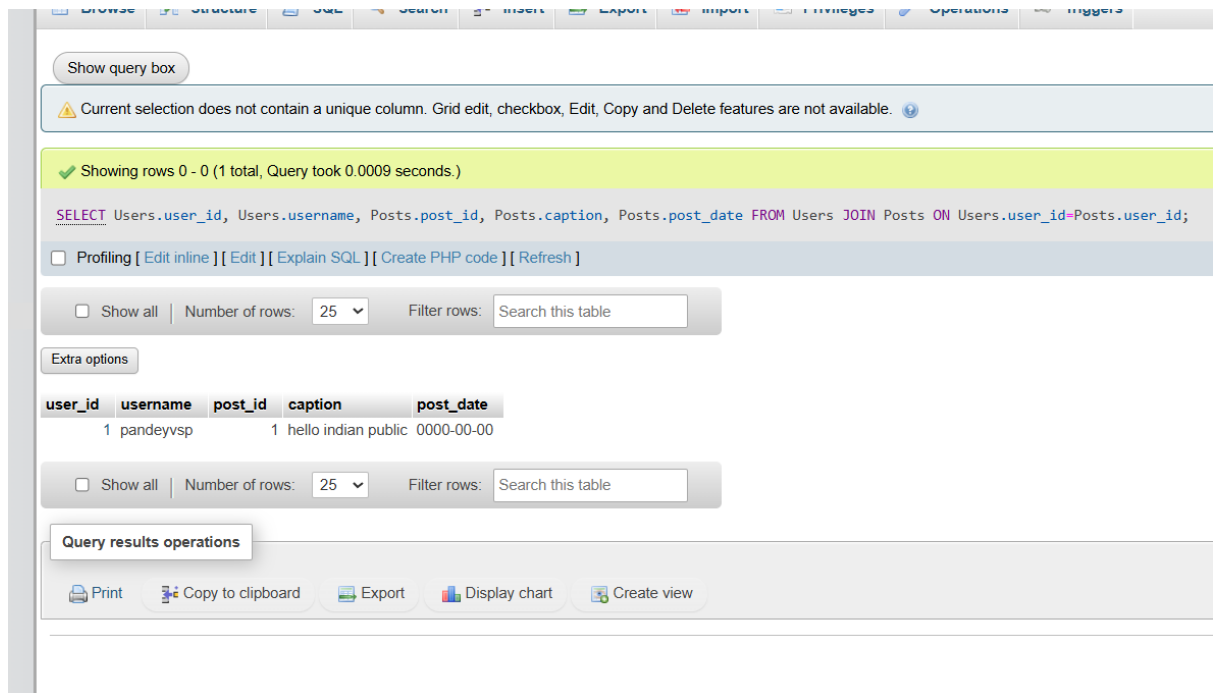
☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

4. Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

Ans.



5. Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

Ans.

MySQL

MySQL is the standard database choice for web applications, content management systems like WordPress, and e-commerce platforms. It balances speed, reliability, and ease of use, making it popular among startups and giant tech platforms alike. This database excels in read-heavy environments where rapid data retrieval is essential.

PostgreSQL

PostgreSQL is a powerful object-relational database favored for complex web applications, financial systems, and data analytics. It offers advanced data types, strict SQL compliance, and robust support for concurrent write operations. Developers choose it when data integrity, extensibility, and complicated queries are primary requirements.

Oracle

Oracle is a high-performance enterprise database designed for massive corporations, banking systems, and large-scale industrial applications. It provides top-tier security, unmatched scalability, and advanced features for handling mission-critical, high-volume transactions. Organizations choose it when they require dedicated vendor support and have the budget for premium licensing.

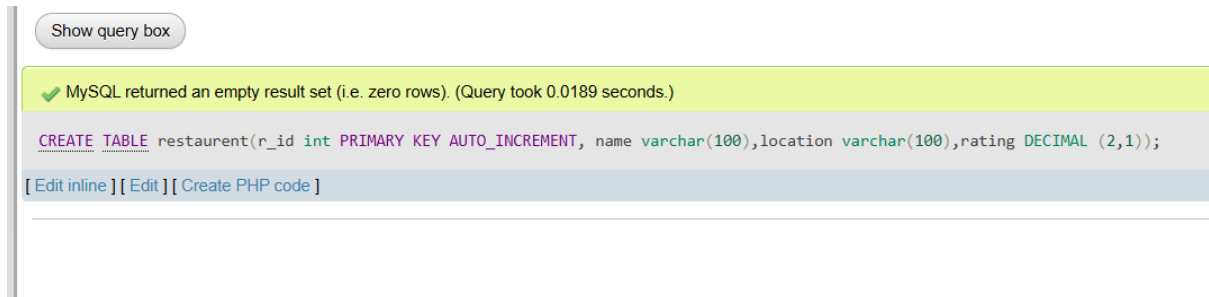
SQLite

SQLite is a lightweight, file-based database built directly into mobile applications, desktop software, and internet-of-things (IoT) devices. It operates without a separate server process, storing the entire database within a single file on the host machine. This makes it perfect for local storage, rapid prototyping, and applications with low-concurrency needs

Session-2

1. Open MySQL Workbench or CLI and write a **CREATE TABLE** statement to create a table called 'restaurants' with columns: id (INT, primary key, auto-increment), name (VARCHAR(100)), location (VARCHAR(100)), and rating (DECIMAL(2,1)).

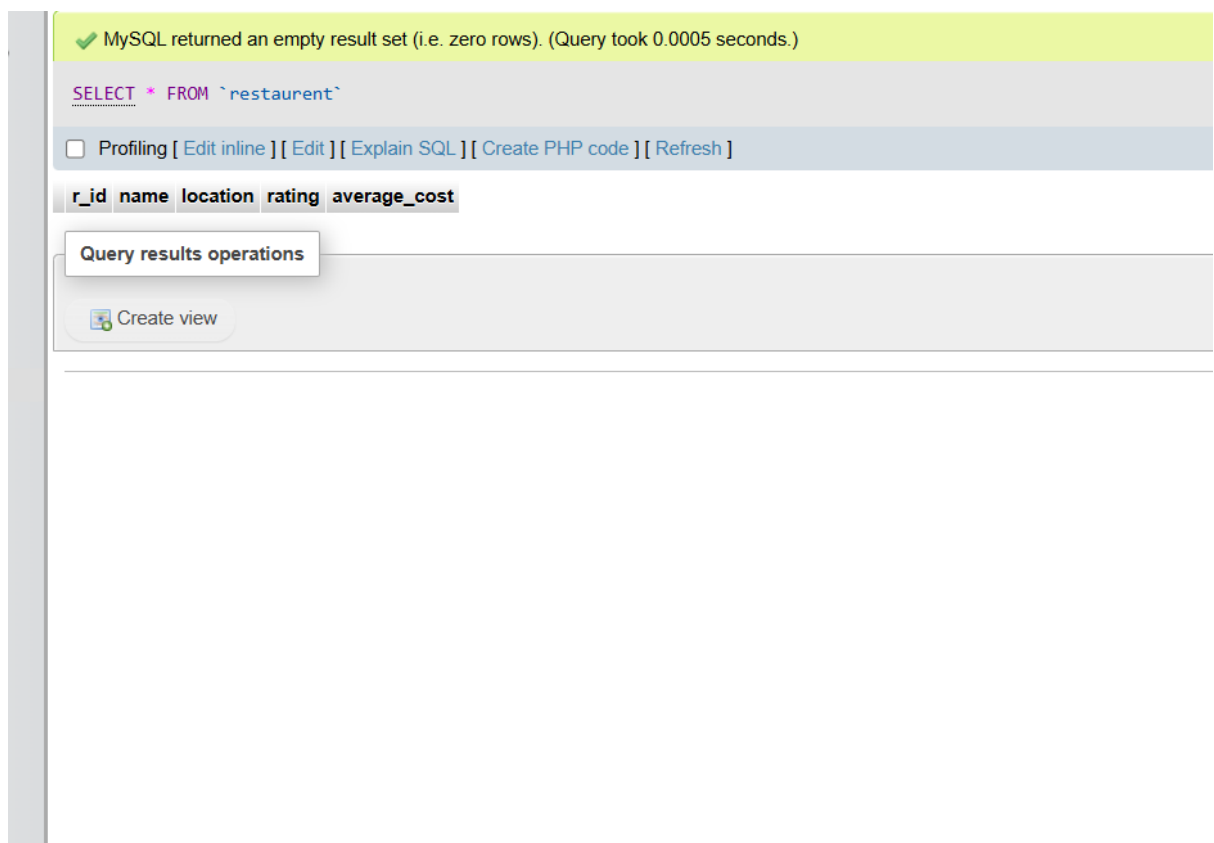
Ans.



2. Use the ALTER TABLE command in MySQL to add a new column 'average_cost' (INT) to your 'restaurants' table.

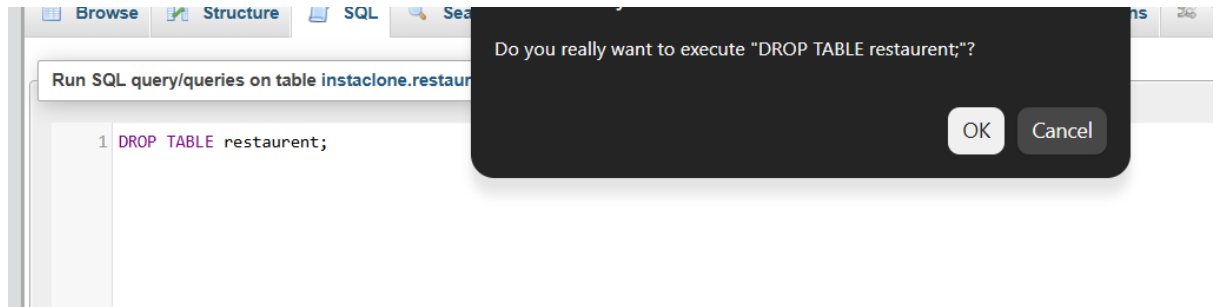
Hint:
Remember to specify the data type when adding a new column.

Ans.



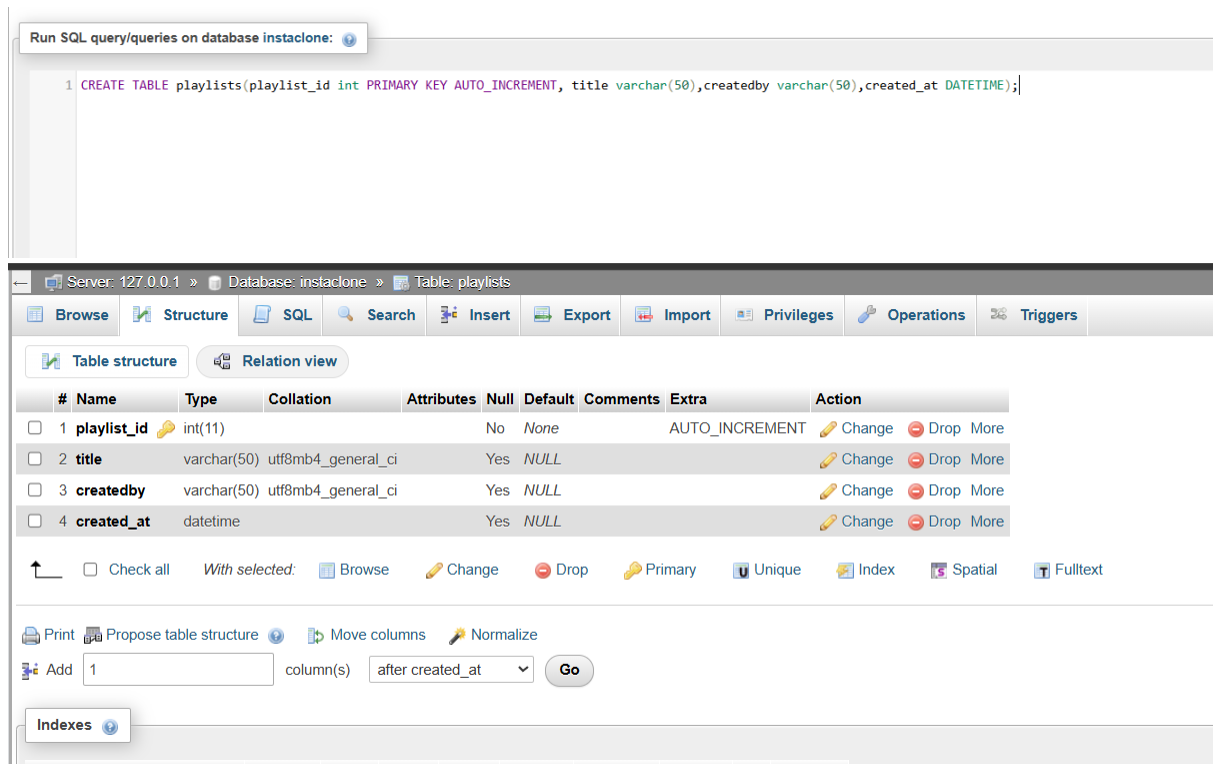
3. Write a SQL statement to DROP the 'restaurants' table you created earlier. Double-check your command before running it to avoid accidental data loss.

Ans.



4. Create a new table called 'playlists' for a music app like Spotify, with columns: playlist_id (INT, primary key, auto-increment), title (VARCHAR(50)), created_by (VARCHAR(50)), and created_at (DATE).

Ans.



Session-3

1. Create a SQL table called Restaurants with columns: id (INT, auto-increment), name (VARCHAR), cuisine (VARCHAR), rating (DECIMAL), and city (VARCHAR). Insert 5 sample restaurants into the table, each with a different cuisine and rating.

Ans.

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

```
SELECT * FROM `restaurent`
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				r_id	name	cuisine	rating	city
☐	Edit	Copy	Delete	1	vikash	indian	1.00	ahmedabad
☐	Edit	Copy	Delete	2	karan	chinise	NULL	vadodara
☐	Edit	Copy	Delete	3	nimesh	pujabi	1.00	dahod
☐	Edit	Copy	Delete	4	jay	jain	1.00	shimla
☐	Edit	Copy	Delete	5	vijay	rajasthani	1.00	jaipur

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

2. Write a SQL UPDATE statement to change the rating of the restaurant named 'Spice Villa' to 4.7 in the Restaurants table.

Ans.

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

`SELECT * FROM `restaurant``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	r_id	name	cuisine	rating	city
☐	1	vikash	indian	1.00	ahmedabad
☐	2	karan	chinise	4.70	vadodara
☐	3	nimesh	pujabi	1.00	dahod
☐	4	jay	jain	1.00	shimla
☐	5	vijay	rajasthani	1.00	jaipur

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

3. Delete all restaurants from the Restaurants table where the rating is less than 3.5 using a SQL DELETE statement.

Ans.

Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

`SELECT * FROM `restaurant``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	r_id	name	cuisine	rating	city
☐	2	karan	chinise	4.70	vadodara

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

4. Use a SELECT query with WHERE and LIMIT to fetch the top 2 restaurants in the city 'Ahmedabad' with a rating above 4.0 from the Restaurants

table.

Hint: Combine WHERE for city and rating, and LIMIT to restrict the number of results.
Ans.

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT * from restaurant WHERE rating>4 AND city="ahmedabad";
```

Profiling

[\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all

Number of rows: 25

Filter rows:

Sort by key: None

Extra options

↩

⌵

→

r_id

name

cuisine

rating

city

☐

6

hyatt

italiyan

4.50

ahmedabad

☐

7

marriot

punjabi

4.80

ahmedabad

☐ Check all

With selected:

Edit

Copy

Delete

Export

☐ Show all

Number of rows: 25

Filter rows:

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Session-4

1.Create a table called Playlist with columns: id, song_name, artist, genre, and play_count. Insert at least 5 sample records representing your favorite songs from Spotify.

Ans.

✓ Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

```
SELECT * FROM `playlist`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				p_id	song_name	artist	genre	play_count
☐	Edit	Copy	Delete	1	teri yaad	pavan	yes	4
☐	Edit	Copy	Delete	2	mohhabat betchata ab bajar me	pavan sing	yes	10
☐	Edit	Copy	Delete	3	aja savan	khesarilal	no	15
☐	Edit	Copy	Delete	4	aaj fir tum pe	kjal raj	no	15
☐	Edit	Copy	Delete	5	so ja kale kava	jagdish	yes	2

⬅️ ➡️ ☐ Check all With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

2.Write a SELECT query to display only the song_name and artist from the Playlist table, but show the artist column as 'Singer' using an alias.

Ans.

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
SELECT song_name,artist AS singer FROM playlist;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	song_name	singer
☐ Edit Copy Delete	teri yaad	pavan
☐ Edit Copy Delete	mohhabat betchata ab bajar me	pavan sing
☐ Edit Copy Delete	aja savan	khesarilal
☐ Edit Copy Delete	aaj fir tum pe	kjal raj
☐ Edit Copy Delete	so ja kale kava	jagdish

↑ ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

3. Write a SELECT query to find all songs in the Playlist table where the genre is 'Pop' and play_count is greater than 100. Sort the results by play_count in descending order.

Ans.

Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
SELECT * FROM playlist WHERE genre="pop" AND play_count>100;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	p_id	song_name	artist	genre	play_count
☐ Edit Copy Delete	1	teri yaad	pavan	pop	200
☐ Edit Copy Delete	3	aja savan	khesarilal	pop	600

↑ ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

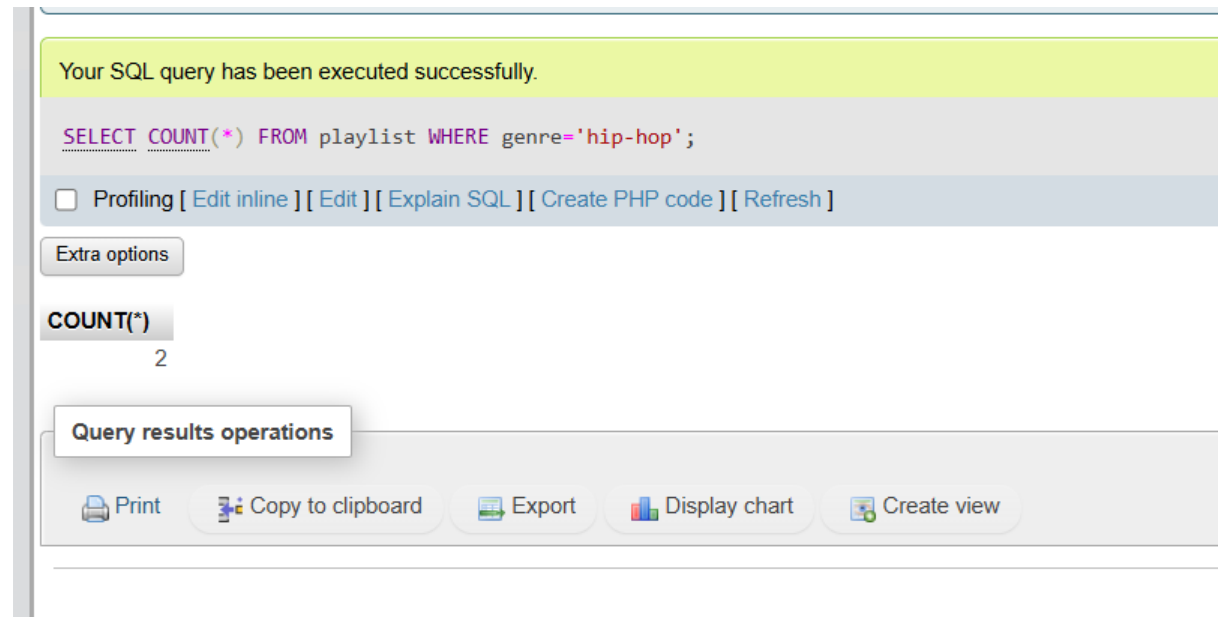
☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

4. Use the COUNT aggregate function to find out how many songs in your Playlist table belong to the genre 'Hip-Hop'.

Ans.



The screenshot shows a web-based SQL interface. At the top, a green message box states "Your SQL query has been executed successfully." Below this, the SQL query is displayed: `SELECT COUNT(*) FROM playlist WHERE genre='hip-hop';`. Underneath the query, there are several links: "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". A button labeled "Extra options" is also present. The query result is shown in a table with one row:

COUNT(*)
2

. At the bottom, there is a "Query results operations" section with buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view".

5.

Write a SELECT query that shows the total number of plays (SUM of play_count) for each genre in your Playlist table, grouped by genre.
Hint: Use GROUP BY with the SUM function to aggregate play counts by genre.

Ans.

Show query box

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT genre, SUM(play_count) AS total_plays FROM Playlist GROUP BY genre;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

 |

Number of rows: 25

 |

Filter rows: Search this table

Extra options

genre

 |

total_plays

☐

Edit

Copy

Delete

hip-hop

1101

☐

Edit

Copy

Delete

pop

900

Check all

 |

With selected:

Edit

Copy

Delete

Export

Show all

 |

Number of rows: 25

 |

Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Session-5

1. Create two tables in SQL: influencers (with influencer_id as PRIMARY KEY and name) and posts (with post_id as PRIMARY KEY, influencer_id as FOREIGN KEY, and caption). Insert at least 3 influencers and 2 posts for each influencer.

Ans.

Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

```
SELECT * FROM `influencers`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	influencers_id	influencers_name
☐ Edit Copy Delete	1	karan
☐ Edit Copy Delete	2	vikash
☐ Edit Copy Delete	3	nimesh

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Showing rows 0 - 1 (2 total, Query took 0.0004 seconds.)

```
SELECT * FROM `post`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	post_id	caption	influencers_id
☐ Edit Copy Delete	1	focus on the good	1
☐ Edit Copy Delete	2	good coders	2

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

2. Write an SQL query using INNER JOIN to display each post's caption along with the name of the influencer who posted it, based on the tables you created.

Ans.

Showing rows 0 - 1 (2 total, Query took 0.0009 seconds.)

```
SELECT * FROM influencers AS i INNER JOIN post AS p ON i.influencers_id=p.post_id;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

influencers_id	influencers_name	post_id	caption	influencers_id
1	karan	1	focus on the good	1
2	vikash	2	good coders	2

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

3. Write an SQL query using LEFT JOIN to show all influencers and their posts, including influencers who haven't posted anything yet. If an influencer has no posts, display 'No Posts' in the caption column.

Hint: Use IFNULL or COALESCE to handle NULL values in the result.

Ans.

Showing rows 0 - 2 (3 total, Query took 0.0019 seconds.)

```
SELECT * FROM influencers AS i LEFT JOIN post AS p ON i.influencers_id=p.post_id;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

influencers_id	influencers_name	post_id	caption	influencers_id
1	karan	1	focus on the good	1
2	vikash	2	good coders	2
3	nimesh	NULL	NULL	NULL

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

4. Write an SQL query using RIGHT JOIN to list all posts and the corresponding influencer's name, ensuring that even posts without a matching influencer_id (if any) are shown.

Ans.

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✓ Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT * FROM influencers AS i RIGHT JOIN post AS p ON i.influencers_id=p.post_id;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Extra options

influencers_id	influencers_name	post_id	caption	influencers_id
1	karan	1	focus on the good	1
2	vikash	2	good coders	2

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Session -6

1. Create a SQL query using GROUP BY to show the total number of orders placed by each user in an 'orders' table, displaying user_id and order_count.

Ans.

The screenshot shows a web-based SQL interface. At the top, there is a 'Show query box' button. Below it, a green status bar indicates 'Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)'. The SQL query is displayed: `SELECT user_id, COUNT(order_id) AS order_count FROM orders GROUP BY user_id;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. An 'Extra options' button is also present. The query results are displayed in a table with two columns: 'user_id' and 'order_count'. The results are as follows:

user_id	order_count
101	3
102	2
103	1

Below the table, there is another control bar with 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. At the bottom, there is a 'Query results operations' section with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

- 2 . Write a SQL query to display the names of restaurants from a 'restaurants' table that have received an average rating above 4.0, using GROUP BY and HAVING.

Ans.

Server: 127.0.0.1 » Database: instadone » Table: restaurant

Run SQL query/queries on table instadone.restaurant:

```

1 SELECT r_id,
2 AVG(rating) AS rt
3 FROM restaurant
4 GROUP BY r_id
5 HAVING AVG (rating)>4.0;

```

Showing rows 0 - 3 (4 total, Query took 0.0006 seconds.)

```

SELECT r_id, AVG(rating) AS rt FROM restaurant GROUP BY r_id HAVING AVG (rating)>4.0;

```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

	r_id	rt
☐ Edit Copy Delete	2	4.700000
☐ Edit Copy Delete	6	4.500000
☐ Edit Copy Delete	7	4.800000
☐ Edit Copy Delete	9	5.000000

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows:

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

3. Suppose you have a 'payments' table with columns user_id, amount, and payment_date. Write a subquery to find user_ids who have made a single payment above ₹2000 in any transaction.

Ans.

✓ Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
SELECT r_id FROM restaurant WHERE payment>2000;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	r_id
☐ Edit Copy Delete	2
☐ Edit Copy Delete	6

↑ ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

4. Using a nested subquery, display the names of movies from a 'movies' table that have a higher average rating than the overall average rating of all movies.

Hint: Use a subquery to calculate the overall average rating, then compare each movie's average rating against it in the main query.

Ans.

Show query box

✓ Showing rows 0 - 3 (4 total, Query took 0.0029 seconds.)

```
SELECT movie FROM restaurant WHERE rating > ( SELECT AVG(rating) FROM restaurant );
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	movie
☐ Edit Copy Delete	jungle
☐ Edit Copy Delete	dhurandhar-2
☐ Edit Copy Delete	rajan

↑ ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Delete](#) [Copy to clipboard](#) [Export](#) [Display sheet](#) [Create view](#)

Session-7

1. Create a table called Restaurant with columns: id, name, location, and cuisine. Apply NOT NULL constraint to name and location, and make id the PRIMARY KEY.

Ans.

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0010 seconds.)

```
SELECT * FROM `restaurants`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

rs_id	rs_name	location	cuisine
-------	---------	----------	---------

Query results operations

[Create view](#)

2. Create a table called FoodOrder with columns: order_id, restaurant_id, user_id, and order_total. Set order_id as PRIMARY KEY, and add a FOREIGN KEY constraint on restaurant_id referencing Restaurant(id).

Ans.

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0089 seconds.)

```
CREATE TABLE foodaorder(order_id int PRIMARY KEY AUTO_INCREMENT,rs_id int, user_id int,order_total int, FOREIGN KEY (rs_id)REFERENCES restaurants(rs_id));
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```
SELECT * FROM `foodaorder`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

order_id	rs_id	user_id	order_total
----------	-------	---------	-------------

Query results operations

[Create view](#)

3. Insert 6 rows into a table called SpotifyUser with columns: user_id, username, email, and subscription_type. Ensure that username is UNIQUE and email is NOT NULL.

Hint: Use the UNIQUE and NOT NULL constraints when creating the table.

Ans.

The screenshot shows a database management interface with a toolbar at the top containing buttons for Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and Triggers. A status bar at the top indicates 'Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)'. Below this, the SQL query 'SELECT * FROM `spotifyuser`' is displayed. A 'Profiling' section with links for 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh' is visible. A control bar allows showing all rows, setting the number of rows to 25, filtering rows, and sorting by key. An 'Extra options' button is present. The main area displays a table with 6 rows and 4 columns: user_id, username, email, and subscription_type. Each row has edit, copy, and delete icons. Below the table, there are options to check all, edit, copy, delete, or export the selected rows. A second control bar is identical to the first. At the bottom, a 'Query results operations' section includes buttons for Print, Copy to clipboard, Export, Display chart, and Create view.

	user_id	username	email	subscription_type
☐	1	alice	ali@gmail.com	premium
☐	2	bobbeat	bob@gmail.com	free
☐	3	charlei	char@gmail.com	premium
☐	4	david	davi@gmail.com	free
☐	5	evasound	eva@gmail.com	free
☐	6	melody	mel@gmail.com	premium

4. Create a VIEW named TopSpendersView that shows usernames and order_total from a FoodOrder table where order_total is greater than 1000.

Ans.

Show query box

✓ Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
SELECT * FROM foodaorder WHERE order_total > 1000;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			order_id	rs_id	user_id	order_total	
☐	Edit	Copy	Delete	6	1	NULL	2500
☐	Edit	Copy	Delete	7	2	NULL	2600

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

SESSION -8

1. Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

Ans.

2. Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.

Ans.

The screenshot shows a database query interface. At the top, there is a 'Show query box' button. Below it, a green status bar indicates 'Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)'. The SQL query is displayed: `SELECT rs_id,order_total FROM foodaorder WHERE order_total>500;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', 'Filter rows: Search this table', and 'Sort by key: None'. An 'Extra options' button is also present. The query results are displayed in a table with columns 'rs_id' and 'order_total'. There are three rows of data. Each row has a checkbox, an 'Edit' button, a 'Copy' button, and a 'Delete' button. Below the table, there are buttons for 'Check all', 'With selected: Edit', 'Copy', 'Delete', and 'Export'. Another control bar is at the bottom of the table area. At the very bottom, there is a 'Query results operations' section with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Show query box

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

`SELECT rs_id,order_total FROM foodaorder WHERE order_total>500;`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	rs_id	order_total
☐ Edit Copy Delete	1	2500
☐ Edit Copy Delete	2	2600
☐ Edit Copy Delete	3	1000

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

3. Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE.
Hint: Use a counter variable and EXIT WHEN condition inside the loop.

Ans.

4. Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE.
Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.

Ans.

The screenshot displays a database query results interface. At the top, a green status bar indicates "Showing rows 0 - 3 (4 total, Query took 0.0008 seconds.)". Below this, the SQL query `SELECT * FROM `products`` is shown. A toolbar contains links for "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". Below the toolbar, there are controls for "Show all", "Number of rows" (set to 25), "Filter rows" (with a search box "Search this table"), and "Sort by key" (set to "None").

An "Extra options" section is visible, followed by a table of results. The table has two columns: `product_id` and `product_name`. It contains four rows of data:

product_id	product_name
1	mobile
2	laptop
3	tablet
4	ipad

Each row has interactive icons for "Edit", "Copy", and "Delete". Below the table, there are checkboxes for "Check all" and "With selected:", followed by "Edit", "Copy", "Delete", and "Export" options. Another set of controls for "Show all", "Number of rows", "Filter rows", and "Sort by key" is present below the table.

At the bottom, a "Query results operations" section includes buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view".

5. Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

Ans.

Session-9

1.Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnecting to the database.

Ans.

The screenshot shows a database management interface. At the top, a green status bar indicates "Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)". Below this, the SQL query `SELECT * FROM `orders`;` is displayed. A toolbar offers options like "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". Below the query, there are controls for "Show all", "Number of rows" (set to 25), and a "Filter rows" search box. An "Extra options" button is also present. The main area displays a table with the following data:

	OrderID	CustomerName	ProductName	Amount
☐	101	Rahul	Laptop	55000.00

Below the table, there are more controls for "Check all", "With selected", "Edit", "Copy", "Delete", and "Export". Another set of "Show all", "Number of rows", and "Filter rows" controls is at the bottom. A "Query results operations" section at the very bottom includes buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view".

2.Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before committing. Check that no new items remain in the table after rollback.

The screenshot shows a database management interface. A green status bar at the top indicates "MySQL returned an empty result set (i.e. zero rows). (Query)". Below this, the SQL query `SELECT * FROM orderitems;` is displayed. A toolbar offers options like "Profiling", "Edit inline", "Edit", "Explain SQL", and "Create PHP". Below the query, there are controls for "Show all", "Number of rows", and a "Filter rows" search box. The main area displays a table with the following headers:

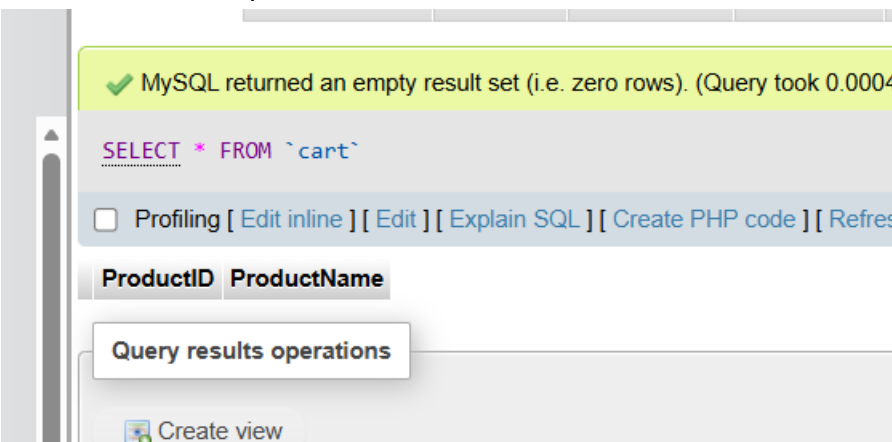
ItemID	FoodName	Quantity
--------	----------	----------

Below the table, there is a "Query results operations" section with a "Create view" button.

Ans.

3. In a Flipkart-like shopping cart scenario, use SAVEPOINT to mark a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.

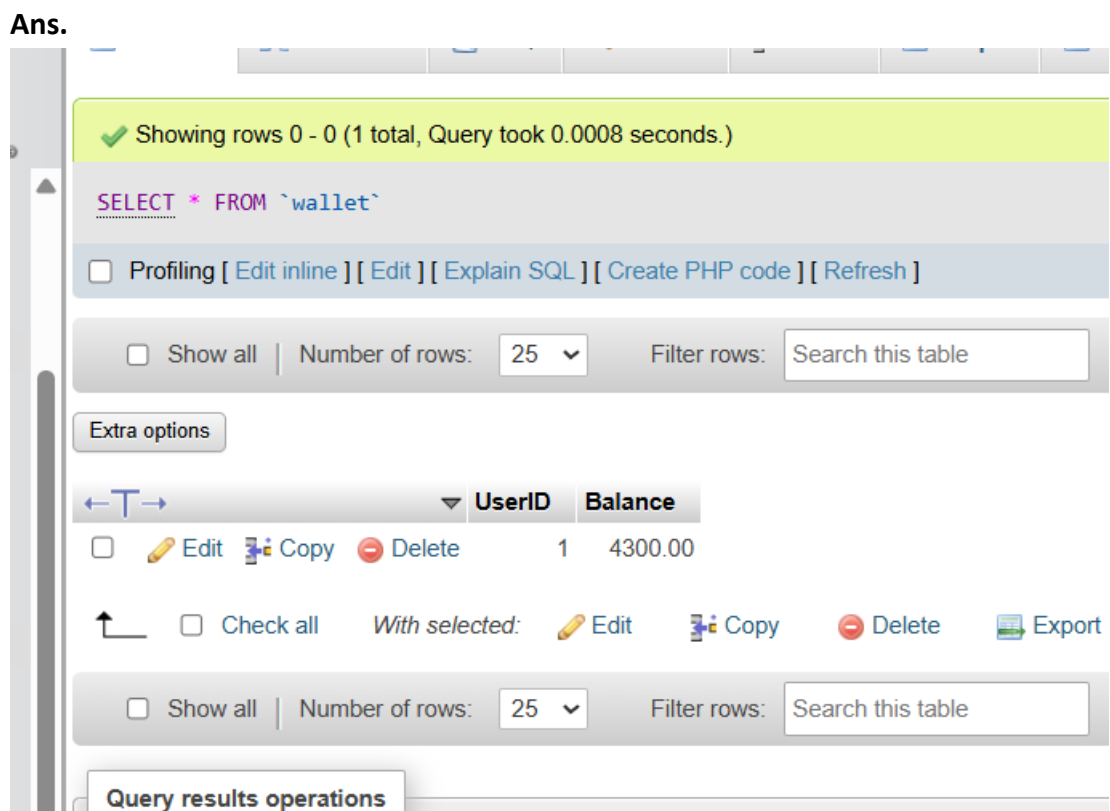
Ans.



The screenshot shows a database interface with a green status bar indicating a successful query: "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)". Below this, the SQL query "SELECT * FROM `cart`" is displayed. A toolbar offers options like "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". The table header shows "ProductID" and "ProductName". A "Query results operations" menu is open, and a "Create view" button is visible at the bottom.

4. Write a trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.

Ans.



The screenshot displays a database query result for the "wallet" table. A green status bar shows: "Showing rows 0 - 0 (1 total, Query took 0.0008 seconds.)". The SQL query "SELECT * FROM `wallet`" is shown. The toolbar includes "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". Below the toolbar, there are controls for "Show all", "Number of rows" (set to 25), and a "Filter rows" search box. An "Extra options" button is also present. The table header shows "UserID" and "Balance". A single row is visible with "UserID" 1 and "Balance" 4300.00. Action buttons for "Edit", "Copy", and "Delete" are provided for this row. At the bottom, there are additional controls for "Check all", "With selected", "Edit", "Copy", "Delete", and "Export". A "Query results operations" menu is open at the bottom.

5. Use ChatGPT to generate a SQL test scenario where a trigger incorrectly updates a user's balance after an expense entry. Paste the scenario and your corrected trigger code, explaining how you fixed the error.

Hint: Ask ChatGPT for a buggy trigger example and a test case that exposes the bug.

Ans.

The screenshot displays a database management tool interface. At the top, a SQL query editor shows a sequence of commands: deleting from the 'Wallet' table, inserting a new row into 'Wallet' with a balance of 1,500, inserting a new row into 'Transactions' with a value of 301,180, and finally selecting all data from the 'Wallet' table. A confirmation dialog box titled 'localhost says' is overlaid on the right, asking if the user wants to execute the 'DELETE FROM Wallet;' command. The dialog lists the subsequent insert and select queries and provides 'OK' and 'Cancel' buttons.

Below the query editor, the execution results are shown in three sections:

- 1 row inserted.** (Query took 0.0016 seconds.)
The executed query is: `INSERT INTO Wallet VALUES (1,5000);`
- 1 row inserted.** (Query took 0.0019 seconds.)
The executed query is: `INSERT INTO Transactions VALUES (301,1,800);`
- Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)**
The executed query is: `SELECT * FROM Wallet;`

The results section for the 'SELECT * FROM Wallet;' query includes a table with the following data:

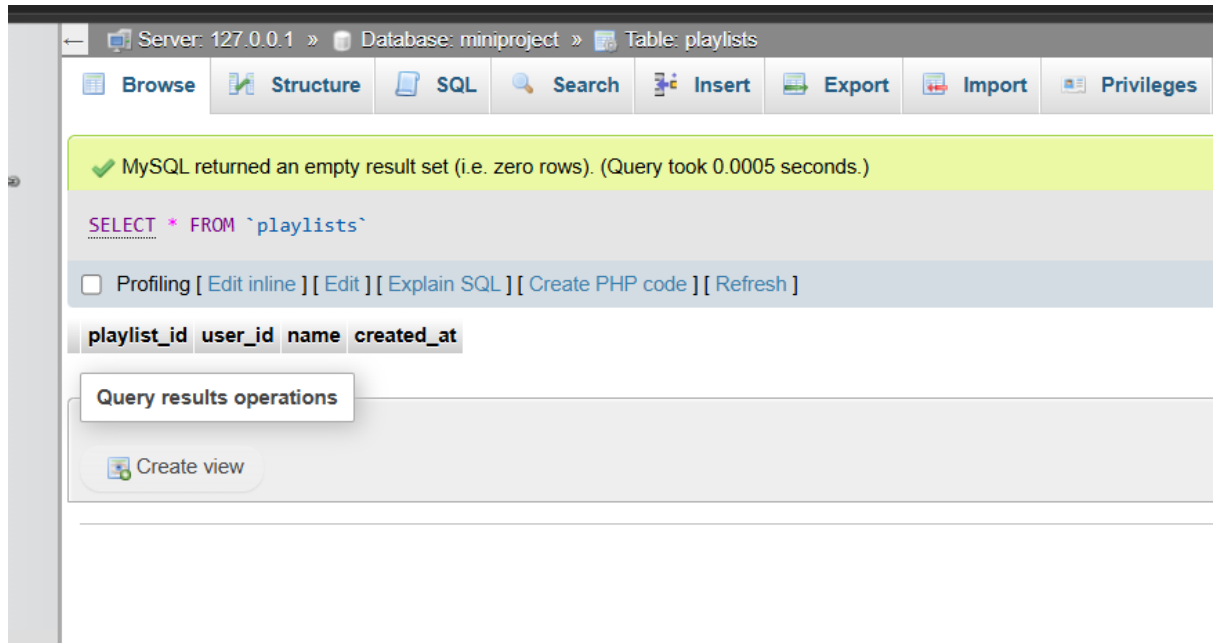
	UserID	Balance
☐ Edit ☐ Copy ☐ Delete	1	0.00

Below the table, there are controls for 'Check all', 'With selected:', 'Edit', 'Copy', 'Delete', and 'Export'. At the bottom, a 'Query results operations' bar contains buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

MiniProject

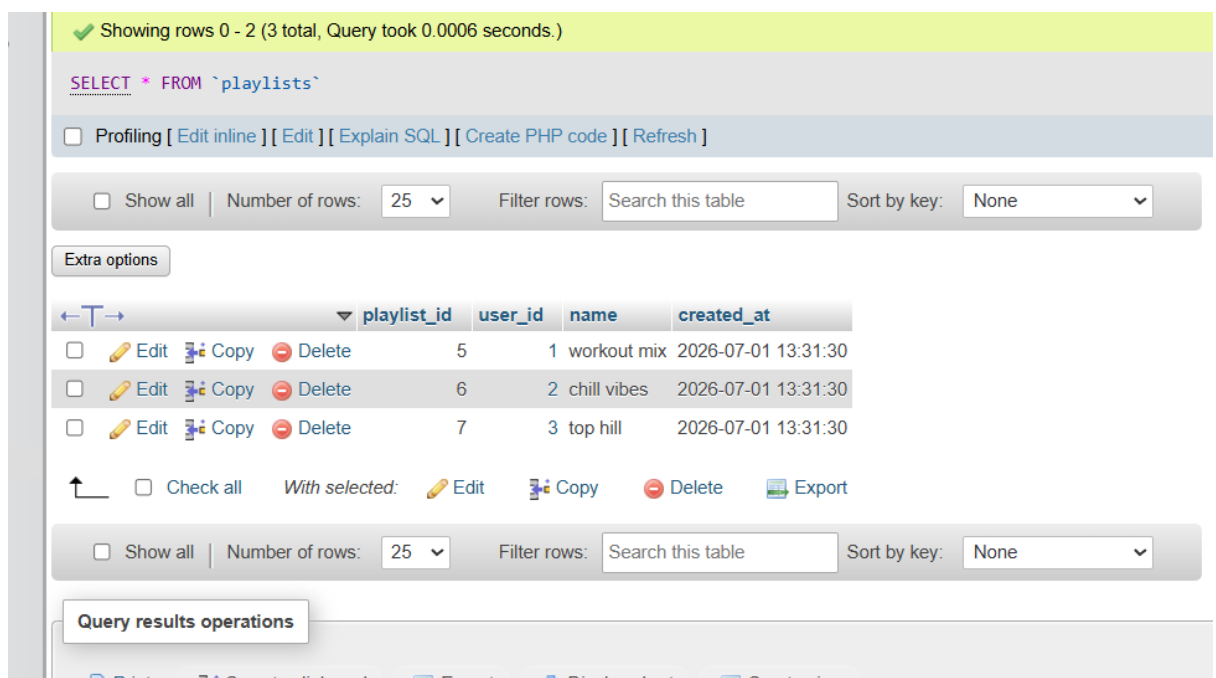
1. Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnecting to the database.

Ans.



2. Insert three sample records into the playlists table: one for a user with a 'Workout Mix', one with a 'Chill Vibes', and one with a 'Top Hits' playlist. Use realistic data for each field.

Ans.



3. Write an SQL UPDATE statement to rename the 'Chill Vibes' playlist to 'Evening Chill' for a specific user_id.

Ans.

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

```
SELECT * FROM `playlists`
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		playlist_id	user_id	name	created_at
☐	Edit		5	1 evening chill	2026-07-01 13:31:30
☐	Edit		6	2 chill vibes	2026-07-01 13:31:30
☐	Edit		7	3 top hill	2026-07-01 13:31:30

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

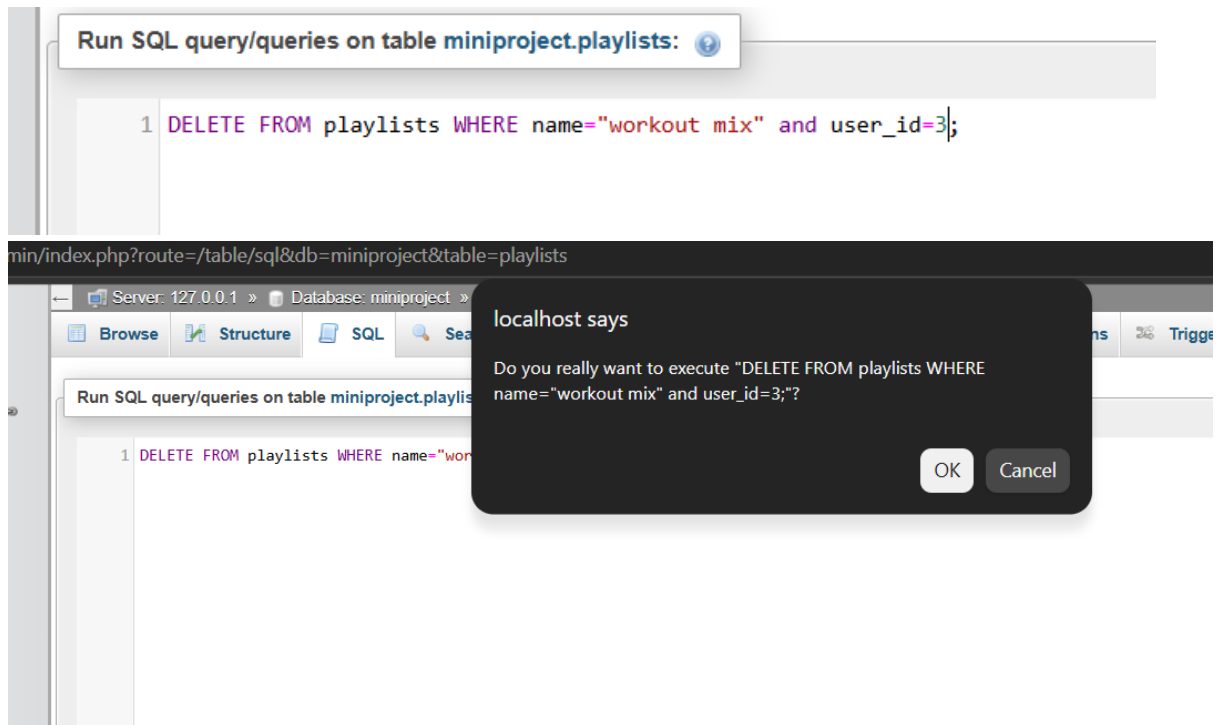
Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

```
1 UPDATE playlists SET name="evening chill" WHERE user_id=1;
```

4. Write a DELETE statement to remove a playlist named 'Workout Mix' for a given user_id from the playlists table.
Hint: Make sure your WHERE clause targets only the intended row.

Ans.



5. Create a stored procedure named GetMonthlyPlaylistCount that takes a user_id and a month (as an integer) as input, and returns the total number of playlists the user created in that month.

Hint: Use the MONTH() function in your WHERE clause.

Ans.

Browse

Structure

SQL

Search

Insert

Export

Run SQL query/queries on table `expense_traker_db.expenses`:

2

3

4

5

6

7

8

9

10

11

12

13

14

15

```
CREATE PROCEDURE GetMonthlyPlaylistCount(  
    IN p_user_id INT,  
    IN p_month INT,  
    OUT p_total_playlists INT  
)  
BEGIN  
    SELECT COUNT(*) INTO p_total_playlists  
    FROM playlists  
    WHERE user_id = p_user_id  
        AND MONTH(created_at) = p_month;  
END $$  
DELIMITER ;
```

You have a previously s
query

SELECT *

SELECT

INSERT

UPDATE

DELETE

Clear

Format